INDIAN OIL CORPORATION LIMITED
(Pipelines Division)
Maintenance, PLHO

IndianOil 2018 YEAR OF TRUST

PLHO/ MNT/MECH/05

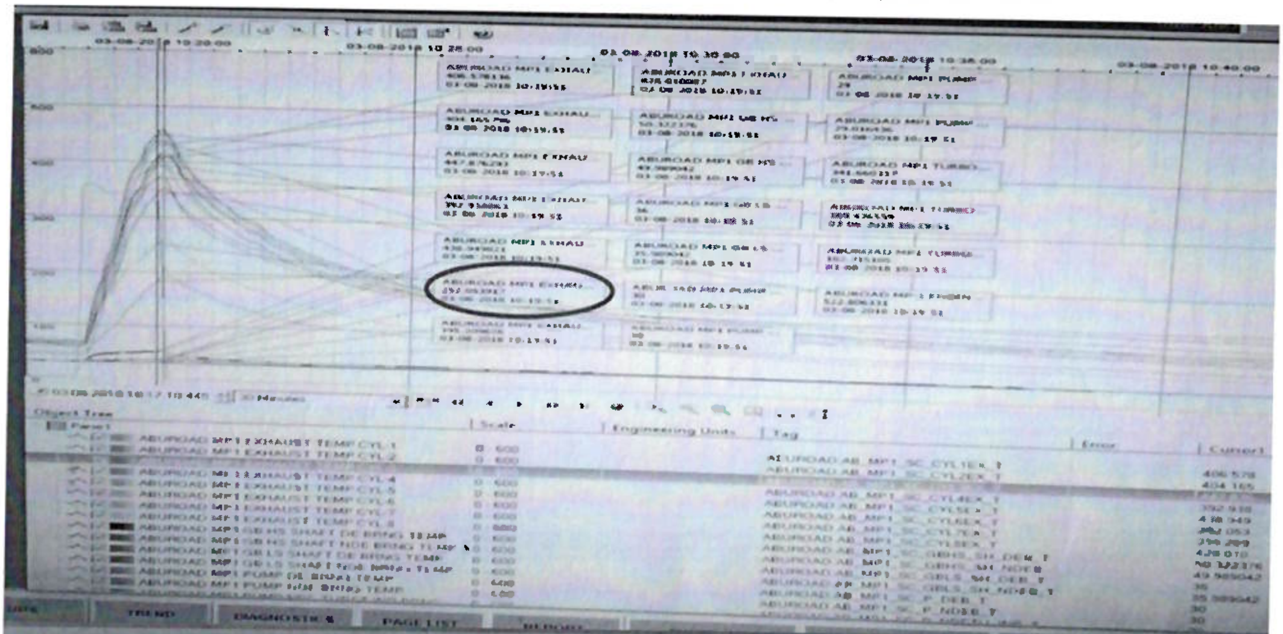
Date: 08.08.2018

Sub: Brief report on breakdown of MP-1 SMPL Abu Road due to failure of cylinder head inlet valve

A. Background

MP1 was started for routine change over at 10:17 Hrs on 03.08.2018 at CRH 98057. The engine had to stop immediately after starting due to abnormal sound observed from engine near cylinder No. 06.

The cylinder exhaust temperature was also very low (250° C) as compared to other cylinders whose temperature was in the range of 400° C. Exhaust temperature trend placed below:



Exhaust Temperature trends of MP-1 at 1017 Hrs on 03.08.2018

Charge air inlet pipe of cylinder No. 06 found heated up while other cylinders charge air Inlet pipes temperature was found normal.

Name Plate Details of MP-1 is placed below:

Engine	Pump
Make- Allen Model- 8S37G Serial No.- D6/50191/16 Rated power- 3146 BHP	Make- BPCL Model- DVDS 14*14*17 ½ Pump RPM- 3240 Pump rated flow rate- 2025 KL/HR Pump Head-334 MCL

B. Brief Maintenance History

Schedule Maintenance Activity:

The last major overhaul (18000 hrs maintenance) of the Engine was done at 96345 hrs and was completed in Jan '2018.

All cylinder heads were replaced with reconditioned cylinder heads during major maintenance along with overhauling/ dynamic balancing of turbocharger.

1500 hrs schedule maintenance was carried out from 16.07.18 to 17.07.18 at 97941hrs. During 1500 Hrs schedule maintenance, tappet clearance was checked and adjusted (if required) for all cylinders.

Recent Maintenance Activity:

On 26.07.2018 at 2310 hrs (CRH 97095), MP-1 tripped in exhaust temperature HI-HI (cyl no. 06) with abrupt increase of Exhaust temperature of all cylinders.

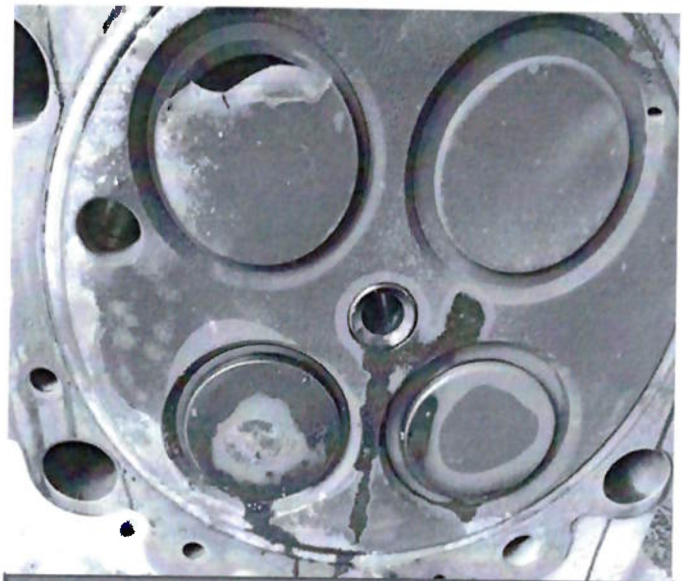
Fuel rack servicing was done and butterfly valve operation checked and found normal. The engine was standby from that day due to operational requirements. The subject failure occurred at cumulative Running hrs of engine of 97095 Hrs i.e. after running of 1712 hrs after last major overhaul.

C. Observation and Maintenance activities done:

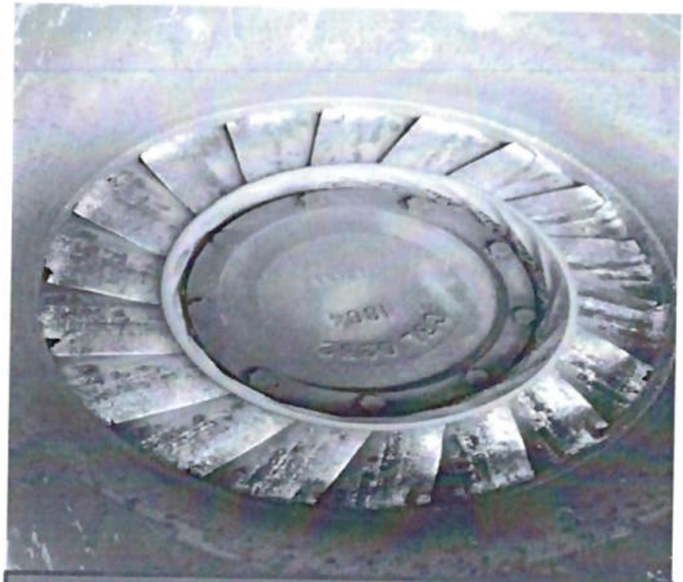
Since the exhaust temperature of cylinder No. 6 was very low and charge air pipe inlet was too hot following activities were carried out:-

1. Tappet clearance checked and found no abnormality.
2. Connecting rod end float was checked along with valve operation by cranking of engine. Everything found normal.
3. To further explore the reasons of abnormal exhaust temperature, it was decided to perform Blow by testing of cylinder No. 06. During Blow by testing, heavy air leakage observed from cylinder head. Therefore cylinder head removed to check out the reasons and it was observed that inlet valve (push rod side) of cylinder head was broken into pieces.
4. The Liner inner surface & piston top surface was also checked and found normal.
5. To find out the broken pieces of inlet valve & further investigate the extent of damage, it was decided to remove double entry of turbocharger. Upon dismantling, stationary nozzle as well as turbocharger turbine blades found damaged which may be due to hitting of broken piece of inlet valve. A broken piece was also found there.
6. Since the broken valve pieces were still not traced, charge air duct of cylinder No. 6 was checked where one small broken piece found.

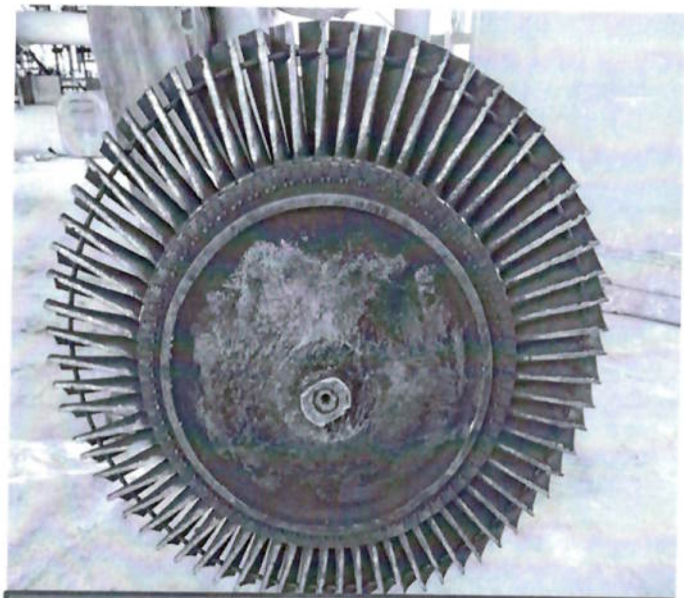
D. **Photographs of Damaged components:**



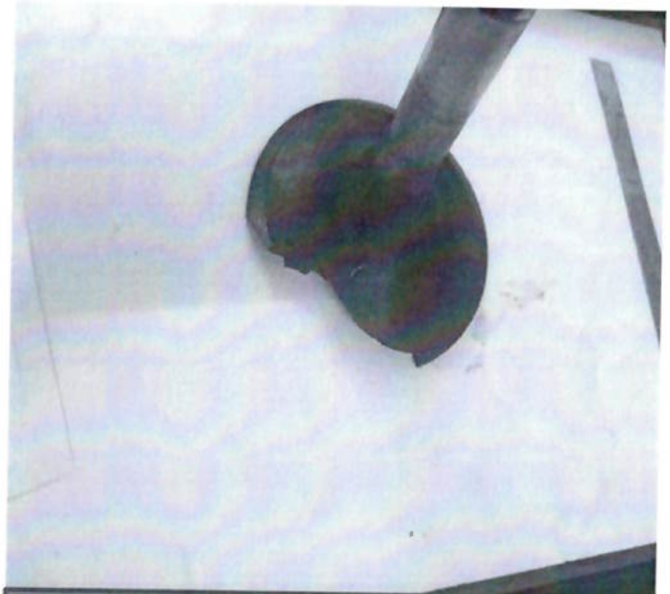
Broken Inlet valve of Cylinder 6



Damaged Stationary Nozzle



Damaged Rotor Assembly



Broken Inlet valve

E. Maintenance activities for restoration

To minimise the downtime of MLPU following maintenance activities are planned.

1. A dynamically balanced spare turbocharger rotor assembly was provided to WRPL AbuRoad since repairing of damaged rotor assembly would take longer time.
2. Turbocharger dismantling completed and preparation of turbocharger rotor assembly is under progress.
3. Reconditioned cylinder head to be fitted parallel to minimise the downtime.
4. As a small piece of broken valve is still not traced, exhaust manifolds are being checked for the same.

F. Probable reasons of failure:

Liner, piston or valve seat does not having any physical damage. Valve tappet clearance was also found within limit. Condition of Valve guide, valve collet and valve rotator will be confirmed only after dismantling of cylinder head.

It is pertinent to mention here that the valve is in service since Sept'2013. As per OEM recommendations, the service life of Inlet Valve is 36000 Hrs. Probable cause of valve failure may be ascertained after testing and failure analysis of inlet valve.

(Amit Kumar)
CMNM, PLHO
08/08/18

~~CGM (M&I)~~

1) What was the service length of failed inlet valve.
2) On 26/7/18 engine had tripped on exhaust temp high alarm. In this context what are the standard checks and whether all these checks were performed. What is the repair sch.

Please speak.

8/8/18

CMNM (AK)

1. Service life of failed inlet valve is approx 18,500 Hrs. Valve is in service since Sep 2013. Service life of Valve is 18000 - 27000 Hrs in case of crude and 36000 Hrs in case of HSD if no visual damage.

PTO

2. following checks were carried out, copy of mail attached:
   - Charge air butterfly valve checked
   - fuel circuit checked for proper fuel supply
   - Temperature prob and SCADA trend also checked

As per the prevailing practice, air circuit & fuel circuit to be checked. These checks were ~~f~~ performed before 03.08.2018. ~~On 03.08.2~~

3. MLPU is expected to be ready for trial run

[illegible]

Copy of e-mail received from WRTL is attached.

अमित
09/08/2018
(Amit Kumar)

~~CGM(M&I)~~

The life of the exhaust valve as per Manual is 36000 hr in diesel. In this case the approx. life was 18,500 hrs.

valve is being sent to TCA, Vadodara for failure analysis.

Matter is also being taken up with M/s Allen Diesel.

A separate guideline is being issued for checks to be performed in case of engine trips on high exhaust temperature.

[signature]
09/8/18

EDFOS

[signature]
13/8

~~CGM(M&I)~~ CMNM (AK)

[signature]
14/8/18

Kumar, Amit (कुमार, अमित)

From: Gupta, Manoj (,)
Sent: 09 August 2018 14:59
To: Kumar, Amit (,)
Cc: Banerjee, D. K (, . .); GUPTA, S.S. (PLHO) (. .); SHENAI, M. R.(. .); PRAJAPATI, RAKESH ( )
Subject: FW: - 08.08.2018

Dear Amit,

1. **Service Length of failed Valve**

As per available records, SAP movement and stamping of valve, this valve is used in Cylinder head reconditioning/refurbishment in Sept'2013 and it was sent to Viramgam in Oct'2013. Considering the average running hours per year as 4000 , the service life rendered before failure is approximately 4000 X 5 = 20,000 . Actual life is slightly lesser considering two 9000 hours maintenance in between , after correcting which, it comes out to be approx. 18500hours.

As per service bulletin of M/s Allen diesel (service life of valve is 27000 in case of crude and 36000 in case of HSD if no visual damage (dent/scores/metal removal) is observed.

Type of Application	Inspect	Replace	Notes
Diesel application	12000 Hrs	36000 Hrs	Dependent upon individual damage i.e. f
HFO/ Crude application	9000 Hrs	18000 – 27000 hrs	

Remarks :- This inlet valve was inspected during last major overhaul (18000 hrs maintenance) of the AbuRoad MLPU which was done in Jan '2018. Condition was ok and therefore re-used.

We are sending inlet valve for metallurgical analysis to TCR Vadodara for ascertaining failure analysis. Also, the matter is being taken up with M/s Allen Diesels as the service life was not exhausted and moreover the type of damage (i.e. normal valve tappet, no hitting marks on piston, no major damage on valve seats, spring found intact and failure happened suddenly) indicates probably material failure.

2. **Service checks carried out after 26.07.2018 when engine tripped in high exhaust temperature**

As per prevailing practice, in case the high exhaust temperature is observed, following checks are performed :-

a) Charge air butterfly to be checked or its open/partial open/closure status.
b) Air in fuel (HSD) supply checked through end plugs
c) Fuel pump rack , its positioning and lubrication to rack
d) Temperature probe/sensor damage/looseness
e) SCADA trend of parameters etc

These checks were performed before trial run on 03.08.2018. For checking other parameters etc, trial run was being taken on 03.08.2018 when engine was stopped immediately (within 1-2 minute) after observing abnormal sound.

3. **Timeline for repair**

Turbocharger assembly made ready. Reconditioned heads fitted. Exhaust piping being fitted back. MLPU is expected to be ready for trial run on or before 11.08.2018.

Once we receive the metallurgical test report from TCR Vadodara and response from M/s Allen Diesels, further details will be supplemented along with these reports.

सादर (Regards)

मनोज गुप्ता | Manoj Gupta
उप महाप्रबन्धक (अनुरक्षण) | DGM (Maintenance)
पश्चिमी क्षेत्र पाइपलाइन्स | Western Region Pipelines
गवरीदड़ (Gauridad) | OFC - 3450

From: Kumar, Amit (कुमार, अमित)
Sent: 08 August 2018 18:22
To: Gupta, Manoj (गुप्ता, मनोज)
Cc: GUPTA, S.S. (PLHO) (एस.एस. गुप्ता); Banerjee, D. K (बनर्जी, डी. के.); SHENAI, M. R.(एम.आर. शनई)
Subject: RE: दैनिक अनुरक्षण रिपोर्ट - 08.08.2018
Importance: High

Dear Sir,

This is in reference to the MP1 SMPL Abu Road inlet valve failure. Following are the observations:

[redacted]

3. What is the time line for repair.

Regards

Amit Kumar
CMNM, PLHO

From: PRAJAPATI, RAKESH (राकेश प्रजापति)
Sent: 08 August 2018 10:10
To: Kumar, Amit (कुमार, अमित)
Cc: SATI, V. C. (सती, वी. सी.); Banerjee, D. K (बनर्जी, डी. के.); Choudhary, K. K. (चौधरी, के. के.); GUPTA, S.S. (PLHO) (एस.एस. गुप्ता); Kamti, Bachcha(कामती, बच्चा); SHENAI, M. R.(एम.आर. शनई); Botke, M (मनीष बोटके); Ghosh Chinmoy (चिन्मय घोष); GUPTA, B K; SAHOO, ALOK; SAWHNEY, S.P. (साहनी, एस.पी.); Sinha, Pankaj K. (WRPL)(पंकज कुमार सिन्हा); Yohan Kumar G (योहान कुमार जी.); YOGESH, VIJAY (योगेश, विजय); Gupta, Manoj (गुप्ता, मनोज); SMPL, Dispatch (एसएमपीएल, डिस्पैच); Parmar, Atul (परमार, अतुल); SHARMA, RAJEEV (राजीव शर्मा); Shankar, V. (शंकर, वी.); Kumar, Neeraj (कुमार, नीरज); SHARMA, ARUN (शर्मा, अरुण); Thakur, Raju (राजू ठाकुर)
Subject: दैनिक अनुरक्षण रिपोर्ट - 08.08.2018

महोदय,

आज की दैनिक अनुरक्षण रिपोर्ट संलग्न है।

सादर (Regards)

ले. राकेश प्रजापति| Lt Rakesh Prajapati
अनुरक्षण प्रबन्धक | Maintenance Manager
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